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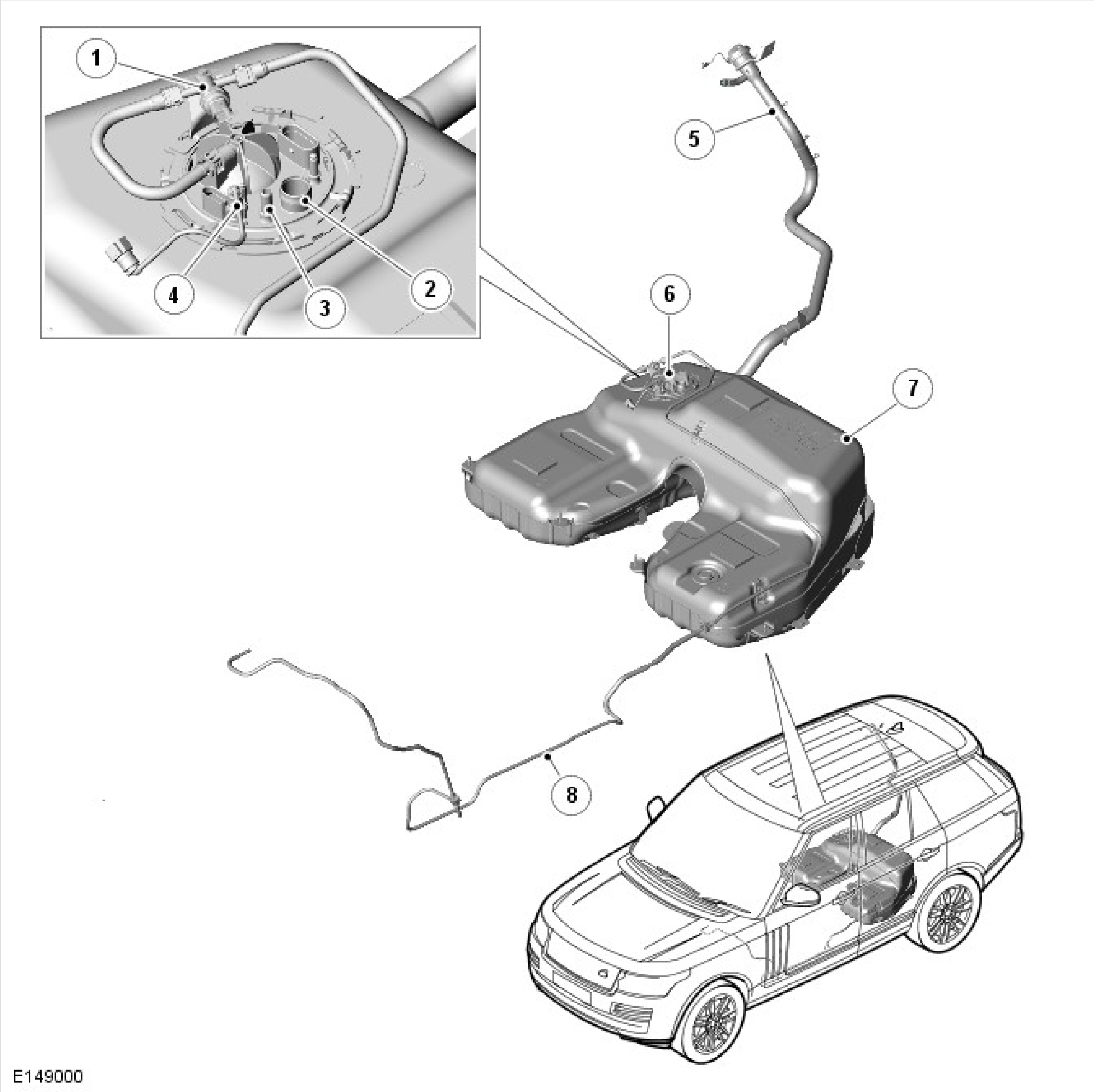
2014.0 RANGE ROVER (LG), 310-01

FUEL TANK AND LINES - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

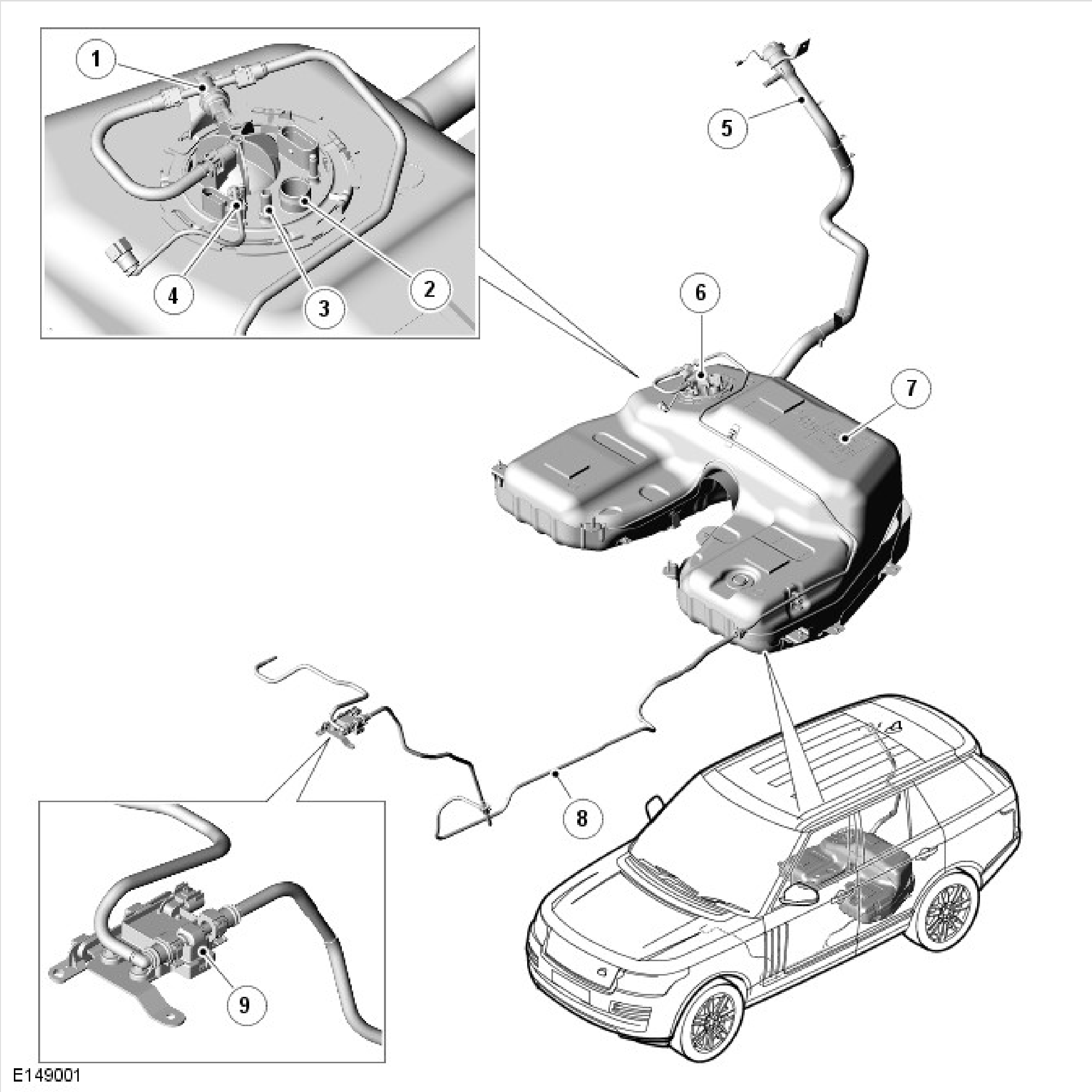
V8 5.0L - FUEL TANK AND LINES COMPONENT LOCATION (ALL EXCEPT NAS)



ITEM	DESCRIPTION
1	Fuel Low Pressure Sensor
2	Fuel Tank Refuelling Breather Connection
3	Fuel Tank Ventilation Pipe Connection
4	Fuel Fired Booster Heater Pipe Connection
5	Fuel Filler Pipe
6	Low Pressure Fuel Pump Module

7	Fuel Tank
8	Fuel Supply Line

V8 5.0L - FUEL TANK AND LINES COMPONENT LOCATION (NAS ONLY)



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8	Fuel Supply Line
9	Fuel Flex Sensor

INTRODUCTION

The V8 5.0L N/A (naturally aspirated) and S/C (Supercharged) low pressure fuel system is designed to supply fuel to the high pressure fuel pumps, mounted on the engine.

The fuel system uses an electronically controlled return less fuel system; this will allow the low pressure pump to deliver fuel to the high pressure pumps, dependant on engine demand. The in-tank low pressure pump's pressure and flow is varied by the engine control module (ECM), with increased system pressure up to 7.3 bar absolute at high temperature and matches the flow in support of engine fuelling requirements.

The V8 5.0L N/A (naturally aspirated) and S/C (Supercharged) fuel pump operation is regulated by a FPDM (fuel pump driver module) which is controlled by the engine management system. The driver module regulates the flow and pressure supplied by controlling the operation of the fuel pump using a PWM (pulse width modulation) output.

FUEL TANK

The fuel tank has a capacity of 105 litres (27.7 US Gallons).

Inside the fuel tank, you will find the location of the low pressure fuel pump module assembly.

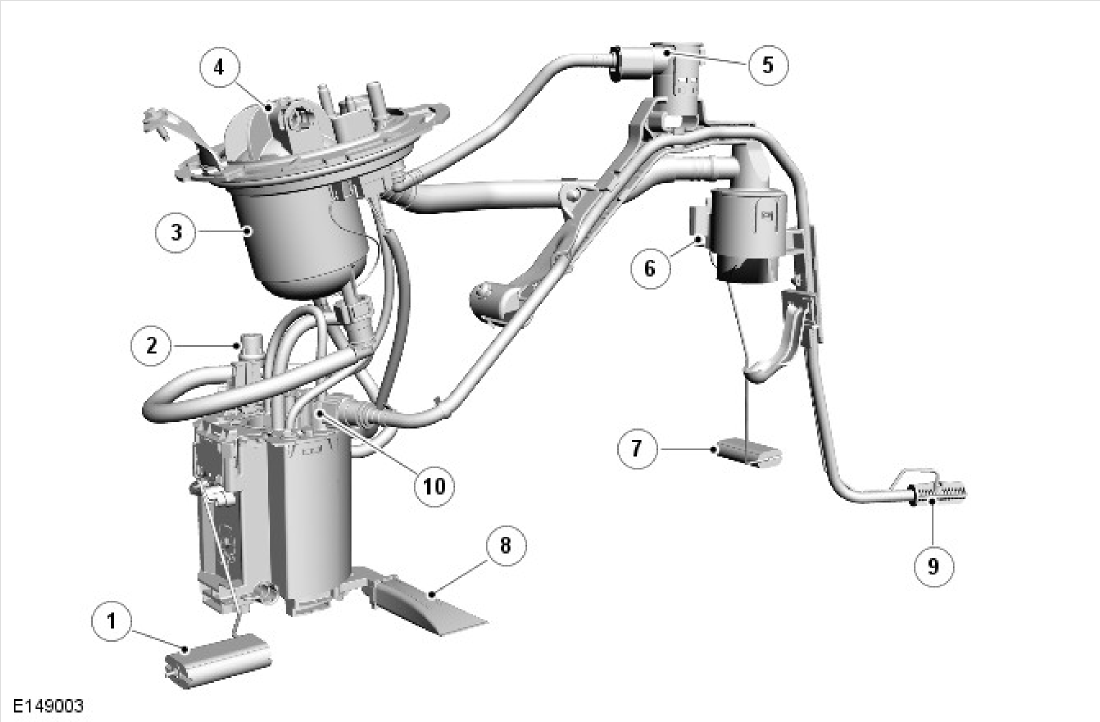
Access to the fuel tank is via the removal of the rear seat base and removal of the low pressure pump module flange located on the right side of the tank.

Within the fuel tank, in the upper left side centre section, a Fill Limit Vent Valve (FLVV) is positioned. This is to control the fuel fill volume into the tank. During the refuelling process, air/vapour is displaced (by the fuel entering the tank) and exits the tank via the Fill Limit Vent Valve (FLVV) and is conveyed by the fuel tank refuelling breather pipe to exit the system either by the filler head for the EURO or ROW vehicles and via the charcoal canister for NAS vehicles. During filling, as the tank reaches its full level, the float in the FLVV is closed by the rising fuel and prevents air/vapour passing through to the refuelling breather pipe. The resulting back pressure causes refuelling to stop automatically.

Also within the fuel tank is a vent/gradient and roll over valve (ROV) in the upper centre section, which allows vapour to escape to prevent tank pressurisation in normal vehicle operating modes; however this valve is designed to close to prevent fuel leakage through the vent in the event of a vehicle roll over.

Fuel level inside the tank is monitored by two level sensors that are connected to the instrument cluster.

FUEL PUMP MODULE



ITEM	DESCRIPTION
1	Right Side Fuel Level Sensor Float
2	Pressure Relief Valve
3	Intank Fuel Filter
4	Fuel Pump Module Tank Flange
5	Fuel Tank Roll Over Valve
6	Fuel Fill Limit Vent Valve
7	Left Side Fuel Level Sensor Float
8	Fuel Pump Coarse Filter
9	Suction Port Filter
10	Venturi Transfer Pump

The fuel pump module is controlled by the engine control module (ECM) via the fuel pump relay located in the rear junction box (RJB).

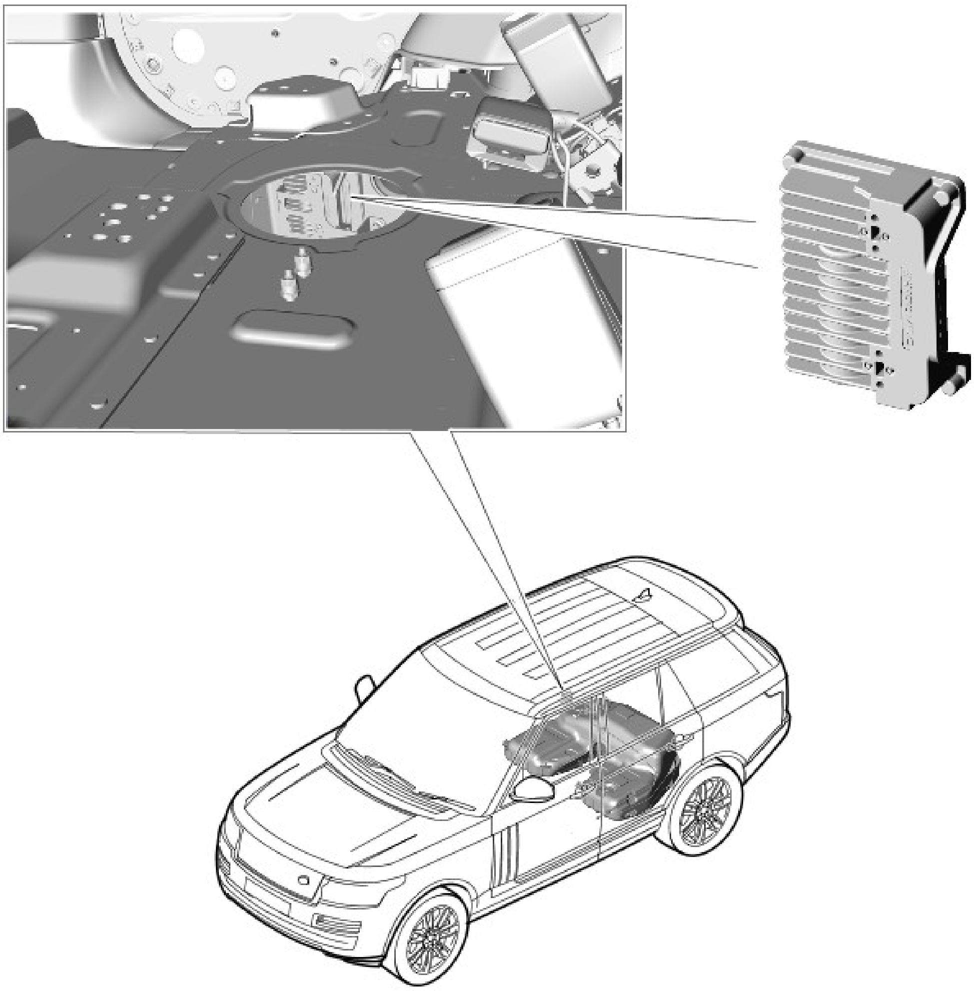
The electric pump collects fuel from the swirl pot and passes it from the tank to the fuel supply line via the fuel filter, to the engine mounted, high pressure pump at a pressure of 5.5 to 7.3 bar.

Fuel is collected from the left side of the tank to the right side of the tank by using a single suction tube that transfers fuel using a venturi transfer pump which is powered by the fuel feed from the fuel pump.

Should the fuel pump electrical connection need to be disconnected, it is important that the ignition is switched off. If the ignition is in position I or II, the instrument cluster will store its last fuel gauge needle position prior to power down. Once power is restored the gauge will display the last stored position regardless of the actual level in the tank.

This could result in incorrect fuel gauge readings if the fuel tank has been drained and not filled with exactly the same quantity of fuel that was removed.

FUEL PUMP DRIVER MODULE (FPDM)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The FPDM (fuel pump driver module) is attached to the right side rear floor member.

The fuel pump operation is regulated by the FPDM which is controlled by the ECM (engine control module). The FPDM regulates the flow and pressure supplied by controlling the operation of the fuel pump using a PWM output.

The FPDM is powered by a supply from the fuel pump relay in RJB (rear junction box). The fuel pump relay is energized on opening the driver’s door or when power mode 9 engine crank is initiates using the stop/start button. The FPDM supplies power to the fuel pump, and adjust the power to control the speed of the fuel pump and thus the pressure and flow in the delivery line.

A PWM signal from the ECM tells the FPDM the required speed for the fuel pump. The on time of the PWM Signal represents half the pump speed, e.g. if the PWM signal with an on time of 50%, the FPDM drives the pump at 100%.

The FPDM will only energize the fuel pump if it receives a valid PWM signal, with an on time of between 4% and 50%. To switch the fuel pump off, the ECM transmits a PWM signal with an on time of 75%.

The output pressure from the fuel pump will change with changes of engine demand and fuel temperature. The ECM monitors the input from the fuel rail low pressure sensor and adjusts the speed of the fuel pump as necessary to maintain an ECM requested output pressure of 5.5 bar to 7.5 bar absolute, during engine start up and engine shut down. Target pressure for the fuel low pressure delivery line is raised to 7.3 bar absolute.

If the Supplemental Restraint System (SRS) outputs a crash signal on the high speed Controller Area Network (CAN) bus, the ECM de-energizes the fuel pump relay to prevent any further fuel being pumped to the engine.

If the ECM does not detect pressure in the fuel delivery line, it stops, or refuses to start the engine and stores the appropriate Diagnostic Trouble Code (DTC).

The ECM receives a monitoring signal from the FPDM. Any DTC's produced by the FPDM are stored by the ECM.

DTC's can be retrieved from the ECM using an approved Land Rover diagnostic system. The FPDM itself cannot be interrogated by the approved Land Rover diagnostic system.

FUEL LEVEL SENSORS

There are two fuel level sensors located inside the fuel tank. One sensor is attached to the swirl pot and monitors fuel level on the right side of the tank. The second is attached on the suction pipe carrier and monitors fuel level on the left side of the tank.

Both level sensors are MAgnetic Passive Position Sensors (MAPPS) which provide a variable resistance to ground for the output from the fuel gauge.

The electrical output signal is proportional to the amount of fuel in the tank and the position of the float arm. The measured resistance is processed by the instrument cluster to implement an anti-slosh function. This monitors the signal and updates the pointer position at regular intervals, preventing constant pointer movement caused by fuel movement in the tank due to cornering and braking.

Fuel Level Sensors - Resistance/Fuel Gauge Read Out Table

NOTE:

These figures are with vehicle on level ground. Sensor readings will defer with varying vehicle inclinations.

Active Side of the Tank:-

SENDER UNIT RESISTANCE, OHMS	NOMINAL GAUGE READING
992	Empty
51	Full

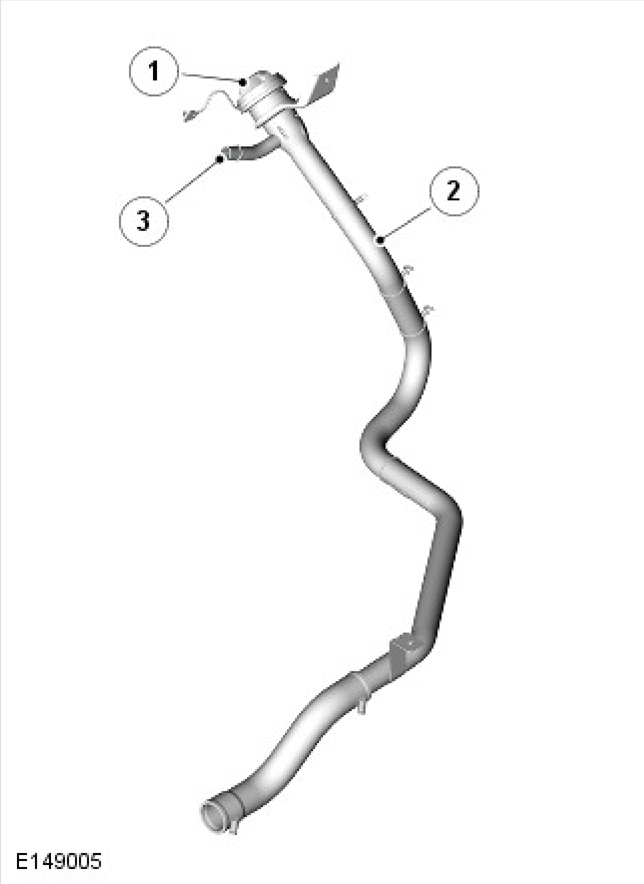
Passive Side of the Tank:-

SENDER UNIT RESISTANCE, OHMS	NOMINAL GAUGE READING
992	Empty
182	Full

When the fuel level becomes low, a fuel warning lamp in the instrument cluster is illuminated and a message is displayed.

The fuel level sender signal is converted into a CAN (controller area network) message by the instrument cluster as a direct interpretation of the fuel tank contents in litres.

FUEL FILLER PIPE



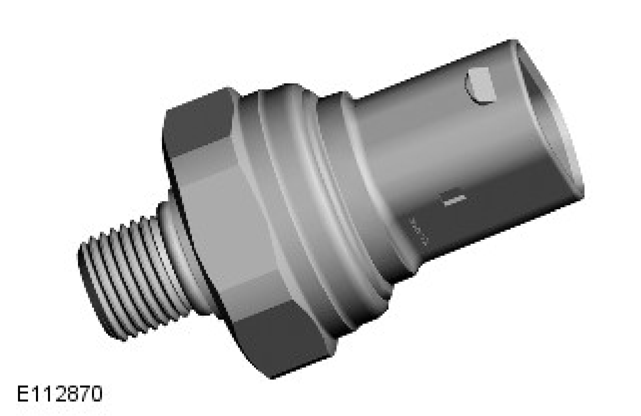
ITEM	DESCRIPTION
1	Fuel Filler Cap
2	Fuel Filler Pipe
3	Refuelling Breather Pipe Connection

The fuel filler pipe locates into the tank and incorporates a spit back flap in the tank end of the pipe. The flap is spring loaded cover which acts as one way valve, allowing the tank to be filled but preventing fuel leaving the tank into the filler pipe.

A connection at the top right side of the filler head allows for the connection of the fuel tank ventilation/breather pipe which goes to the tank. **On NAS vehicles only** , this connection will only be used for EVAP (evaporative emission) System, as the refuelling breather is routed to the charcoal canister.

For additional Information, refer to: Evaporative Emissions (303-13, description and Operation)

FUEL LOW PRESSURE SENSOR



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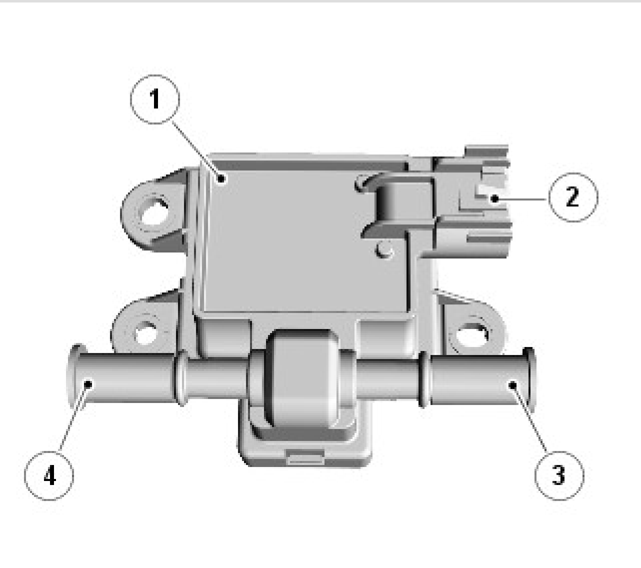
The fuel low pressure sensor supplies a pressure signal to the engine control module (ECM) to enable closed loop control of the fuel pump.

FUEL FLEX SENSOR



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ITEM	DESCRIPTION
1	Fuel Flex Sensor

2	Electrical Connection
3	Fuel Inlet
4	Fuel Outlet

The fuel flex sensor is designed to detect whether the vehicle is operating on ethanol /gasoline fuel and is only **on NAS Vehicles Only** .

The fuel flex sensor is attached to the fuel supply line, and as fuel flows through the sensor, it will determine the ethanol content of the fuel and the temperature of the fuel being delivered to the engine.

The fuel flex sensor is a serviceable part and requires no maintenance.

The output signal from the sensor will be delivered to the engine control module (ECM) in units of frequency (Hz). A pulsated frequency signal (50Hz - 100Hz) corresponds to ethanol percentage (0 - 100%) within an ambient temperature range of 40°C to 125°C.

If measured frequency exceeds that of 170Hz error codes will be displayed.

- Error Codes between 170Hz and 180Hz indicates sensor internal failure and will require replacement.
- Error Code 190Hz indicates fuel composition is outside sensor measurement range and recommend action is to check and replace fuel and retest.